The Keyboard Warrior Effect

Measuring Online Disinhibition in California High-Speed Rail Public Discourse

Introduction to Text Mining and Natural Language Processing
Prof. Hannes Mueller – Barcelona School of Economics

Corneel Moons Sebastian Paik Eric Gutierrez

March 31, 2026

Abstract

We examine the keyboard warrior effect in public discourse around the California High-Speed Rail project, comparing 62,774 comments from YouTube, Reddit, board meetings, and business plan submissions. Using dictionary-based toxicity scoring, two sentiment lexicons, and LDA topic modelling, we find that online comments are significantly more toxic, more negative, and less topically focused than offline comments. A recurring finding is that formal written submissions behave more like in-person board testimony than social media, suggesting that participation effort and accountability drive discourse quality more than the online–offline distinction alone.

Contents

1 Introduction & Motivation **3**

2 Data Collection **4**
2.1 Sources 4
2.2 Collection Pipeline 4

3 Exploratory Data Analysis & Preprocessing **5**
3.1 Corpus Overview 5
3.2 Preprocessing Pipeline 5

4 Analysis **6**
4.1 Research Hypotheses 6
4.2 Methodology 6

5 Results **8**
5.1 H1 – Toxicity 8
5.2 H2 – Negativity Bias 9
5.3 H3 – Relevance 10

6 Conclusion **11**

A Additional Figures for H2 **14**

B Additional Figures for H3 **14**

1 Introduction & Motivation

As the challenges that societies around the globe face get increasingly complex, a trend is emerging that calls for public deliberation as a mechanism to evolve towards more direct, transparent, and citizen-centered institutions, and consequently, towards more direct, transparent, and citizen-centered public policies (Brugúe, 2014, 2022). In this model, the communication among citizens and between citizens and institutions plays a central role, as public policies come into fruition only after a co-creation process where all agents have to say (Parés et al., 2015).

In this context, and as a perversion of the ideal framework posed by the proponents of deliberative institutions, communication channels in both the citizen-citizen axis and the citizen-institutions axis can be misused and abused. When public comments turn away from constructive criticism and respectful disagreement, and towards being outposts for hate, obstruction, and partisan division, the effectiveness and theoretical benefits of deliberative practices fade away. It only follows that a decline in quality of the signals that institutions receive through communication channels can potentially lead to misinformed, opaque, ineffective public policies.

Contemporary to this shift, we have also seen the appearance and the spread of the Internet and social media, which have brought new communication channels to the world (Parés et al., 2015). As institutions increasingly incorporate these new technologies into their decision-making processes, an evaluation of how similar or dissimilar the expression of public sentiment is through these new channels when compared to traditional, offline ones is not only academically rewarding, but also highly relevant.

The theoretical mechanism we draw on is the online disinhibition effect (Suler, 2004): the tendency for individuals to behave with less restraint online than in face-to-face settings, driven by anonymity, invisibility, and the psychological dissociation of the online self. Prior empirical work supports this mechanism in civic contexts. Papacharissi (2004) finds that online political discussion forums frequently fall short of deliberative ideals in terms of civility, despite high levels of participation. Santana (2014) shows that anonymity is a strong predictor of incivility in online news comment boards relative to identified participation.

In this paper, we focus in a particular high-profile, high-spending public policy: the California High-Speed Rail (CAHSR), and we use it as an ideal setting to measure toxicity, negativity, relevance, and topic distribution of the public comments made in four different sources: Youtube comments, Reddit posts, in-person comments during the California High-Speed Rail Authority board meetings, and web-submitted comments to the biennial business plans. Specifically, our goal is to compare and characterize civic engagement across platforms, and between the offline–online divide.

2 Data Collection

2.1 Sources

We construct a corpus of public comments about the California High-Speed Rail project from four distinct channels, spanning the online–offline spectrum of participation.

YouTube and Reddit allow anonymous or pseudonymous participation with no barriers to entry. Board meeting testimony requires physical attendance and identification before a formal audience. Business plan submissions are structured written responses to official CAHSR Authority consultation processes (2020, 2021, 2022, 2024), authored primarily by organisations, advocacy groups, and engaged citizens. This structural variation in participation cost and accountability is the key feature that motivates our online–offline comparison.

Note that we classify the comments to the business plans as offline, provided that even though these are web-submitted, there is no place for user interaction in this source.

2.2 Collection Pipeline

YouTube: Comments were collected using `yt-dlp`[1], an open-source YouTube extraction tool, applied to all publicly accessible videos on the official CAHSR Authority channel. Both top-level comments and full reply threads were retrieved. Comment timestamps, author handles, and like counts were retained.

Reddit: Comments were collected from the `r/cahsr` subreddit using the public Reddit JSON API[2], covering all available posts and their comment threads. Author handles, upvote scores, and thread structure were retained. Note that the Reddit public JSON API paginates results with a practical ceiling of approximately 1,000 posts per listing endpoint; our collection therefore represents a near-complete snapshot of the subreddit at collection time rather than a guaranteed exhaustive archive.

Board Meetings: Public comment transcripts were downloaded as PDFs from the official CAHSR Authority website, covering board meetings from 2018 to 2025. Text was extracted using `pdfplumber`; pages with fewer than 50 characters of extracted text (indicating scanned rather than digitally-created PDFs) were processed with Tesseract OCR as a fallback. Individual speaker comments were segmented from the transcripts and attributed where speaker names were available.

Business Plan: Written public comments were scraped from four CAHSR business plan consultation pages (2020–2024). Where comments referenced attached PDF letters, the letter text was extracted and appended to the submission text. Submission metadata including method (email, website, letter) and submitter name were retained.

All four sources were unified into a single dataset with a common 12-column schema covering: comment text, source, date, author, engagement (likes or upvotes where available, 0 for offline sources), reply flag, source type, word count, and platform-specific identifiers. The exclusion flag column was retained for quality review; no records required exclusion in the current collection, yielding a final analysis corpus of 62,774 comments.

[1] Available at: `https://github.com/yt-dlp/yt-dlp`

[2] Available at: `https://www.reddit.com/dev/api/`

3 Exploratory Data Analysis & Preprocessing

3.1 Corpus Overview

Table 1 presents key descriptive statistics for each source. Notice how the corpus is markedly asymmetric, and the mean length of the comments differs among channels. Online sources account for 98.5% of all comments, reflecting the near-zero participation barriers of social media relative to formal offline channels. Board meeting comments are by far the more extense in length (median of 118 words), consistent with the formal setting requiring prepared, substantive remarks. YouTube comments are the shortest (median of 19 words), reflecting the reactive, conversational nature of social media engagement.

Table 1: Corpus summary statistics by source

Source	Type	N	Share	Mean words	Median words
Board Meetings	Offline	635	1.0%	155	1118
Business Plan	Offline	301	0.5%	84	42
Reddit	Online	24,833	39.6%	52	30
YouTube	Online	37,005	58.9%	35	19
Total		62,774	100%		

Reply structure also differs substantially across platforms. On Reddit, 72.4% of comments are replies to other comments, and on YouTube 64.3% are replies, reflecting active threaded discussion. Offline sources contain no reply structure by definition, as board meeting and business plan comments are addressed directly to the authority rather than to other participants.

3.2 Preprocessing Pipeline

All text is processed through a standard Natural Language Processing (NLP) pipeline prior to vocabulary-based analyses. The steps are applied in the following order:

1. **Lowercasing;** all text is converted to lowercase.
2. **URL and noise removal;** URLs and non-alphabetic characters are removed.
3. **Stopword removal;** standard English stopwords (NLTK) are removed, and tokens shorter than three characters discarded.
4. **Lemmatization;** tokens are lemmatized using the WordNet lemmatizer.

Note that subsequent dictionary-based analyses (H1 toxicity, H2 sentiment) use raw tokenization rather than the full preprocessing pipeline, preserving the original vocabulary for lexicon matching. Lemmatization and stopword removal are applied only for the remaining analyses (H3 constructiveness).

4 Analysis

4.1 Research Hypotheses

We test three hypotheses, each one comparing online sources (YouTube, Reddit) against offline sources (board meetings, business plan submissions) along a different dimension of discourse.

H1 (Toxicity): Online comments exhibit significantly higher toxicity than offline comments, reflecting the reduced social accountability of anonymous or pseudonymous platforms.

H2 (Negativity): Online comments display stronger negative sentiment than offline comments, consistent with the tendency for disinhibited environments to amplify critical expression.

H3 (Relevance): Online comments are less topically relevant to CAHSR than offline comments, serving as a proxy for lower constructiveness.

In all three cases, board meeting comments serve as the primary offline baseline, given their formal institutional context and the requirement for physical attendance and identification. Business plan submissions are included as a secondary offline comparison to test whether the online–offline distinction holds across different types of formal participation.

4.2 Methodology

H1 – Toxicity. To measure toxicity, we utilize the widely adopted Carnegie Mellon University (CMU) Offensive Word List[3], a comprehensive, native-English computational lexicon of profanity and offensive terms. For each comment, toxicity is scored as the proportion of raw tokens matching the CMU dictionary:

$$\text{toxicity} = \frac{\text{toxic token count}}{\text{total token count}}$$

As we hinted earlier, raw tokenization is used to preserve the full vocabulary for lexicon matching, as lemmatization and stopword removal would remove or alter some hostile terms.

H2 – Sentiment Analysis. Sentiment is measured using two established lexicons applied independently as a cross-validation strategy. The AFINN lexicon (Nielsen, 2011) assigns integer scores (-5 to $+5$) to 2,477 English words; we compute a normalised score as the sum of matched word scores divided by total word count. The Bing Liu opinion lexicon (Hu and Liu, 2004) classifies 6,789 words as positive (2,006) or negative (4,783). For comments with at least one matched word, we compute a compound score as:

$$\text{negativity} = \frac{\text{positive words} - \text{negative words}}{\text{positive words} + \text{negative words}}$$

Finally, we assign a sentiment label (positive / neutral / negative) based on the sign of this score. The Bing Liu lexicon operates on raw tokens; the AFINN lexicon matches

[3]Available at: `https://github.com/LDNOOBW/List-of-Dirty-Naughty-Obscene-and-Otherwise-Bad-Words`

against the original (uncleaned) comment text, since its entries include multi-word and punctuated forms. By using two distinct lexicons, and if the results are consistent, a higher level of credibility is achieved, provided that results do not come from a single measurement instrument.

H3 – Relevance. Topical relevance to CAHSR is measured via cosine similarity between each comment and a set of seven project-specific anchor sentences covering construction progress, funding, route alignment, station locations, and project timeline. A document-term matrix (DTM) with trigrams (min_df = 0.001, max_df = 0.85) is fitted on the full preprocessed corpus using a `CountVectorizer`; anchors are transformed into the same space and cosine similarity is computed against each comment vector. The off-topic score is defined as $1 - \max$ cosine similarity to anchors, so that higher values indicate less topically focused comments. Using raw term counts rather than TF-IDF weighting avoids penalising high-frequency CAHSR vocabulary, terms like *rail*, *construction*, and *funding* that appear corpus-wide are precisely the signal we wish to detect. To complement this, we fit a Latent Dirichlet Allocation model (Blei et al., 2003) with $k = 7$ topics using Gensim's `LdaMulticore` on the same DTM, providing an unsupervised view of thematic structure across sources. The number of topics ($k = 7$) was selected after comparing coherence scores (C_v) for $k \in \{5, 7, 10, 15\}$; $k = 7$ tied with $k = 5$ for the highest coherence ($C_v = 0.596$) while producing more granular, interpretable topic labels.

Statistical Testing. Given the non-normal distributions and large sample size imbalance across sources, we use non-parametric tests throughout. We apply one-sided Mann-Whitney U tests comparing each online source against the board meeting baseline. Effect sizes are reported as rank-biserial correlations (r), where $|r| > 0.1$ is considered small, $|r| > 0.3$ medium, and $|r| > 0.5$ large (Cohen, 1988). Because we conduct nine primary hypothesis tests (three per hypothesis), we verify that all significant results survive a Bonferroni-corrected threshold of $\alpha = 0.05/9 \approx 0.0056$. All reported p-values fall well below this threshold except where explicitly noted as non-significant.

5 Results

5.1 H1 – Toxicity

As displayed in Figure 1, the direction of the finding is consistent and robust: 1.9% of YouTube comments and 3.0% of Reddit comments contain at least one toxic term, compared to 0.2% of board meeting comments, and 0.3% of comments to the business plans.

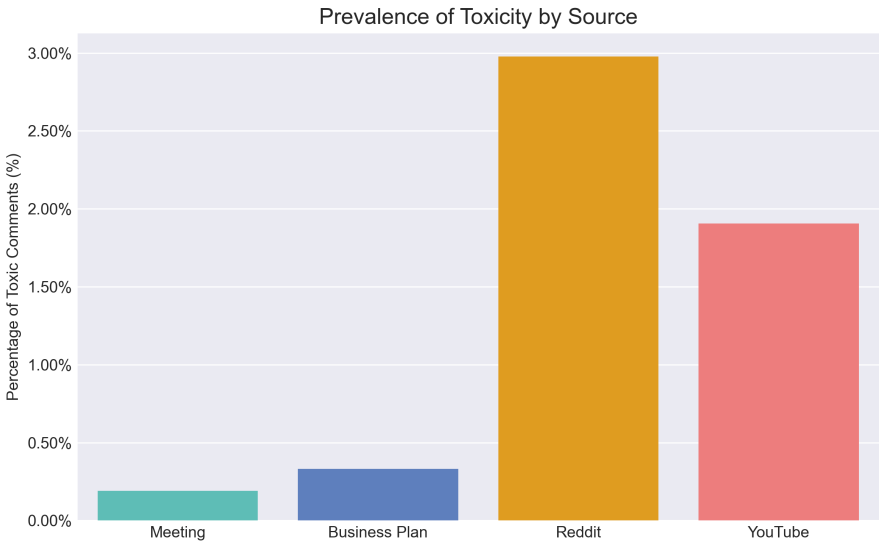


Figure 1: Prevalence of toxicity by source. Online sources show systematically higher toxicity than offline sources.

Table 2 presents Mann-Whitney U test results comparing each source against the board meeting baseline to validate the statistical significance of these differences.

Table 2: H1 – Toxicity: Mann-Whitney U test results

Comparison	Online mean	Meeting mean	Ratio	p-value	Effect size (r)
YouTube vs Meeting	0.0014	0.00004	38.3×	$p = 6.2 \times 10^{-4}$	−0.018
Reddit vs Meeting	0.0014	0.00004	36.2×	$p < 10^{-4}$	−0.028
Business Plan vs Meeting	0.0000	0.00004	1.1×	$p = 0.296$	−0.002

YouTube comments are 38.3 times more toxic than board meeting comments on average, and Reddit comments 36.2 times more toxic. Both differences are statistically significant ($p < 0.01$). The business plan comparison is not statistically significant ($p = 0.296$), suggesting that formal written submissions, despite being submitted online, exhibit toxicity levels closer to in-person board testimony than to social media commentary. This is consistent with the higher participation effort and institutional accountability associated with the business plan process.

Effect sizes are negligible ($r < 0.03$) across all comparisons, falling below Cohen's threshold for a small effect ($|r| > 0.1$). This is an expected property of the dictionary method,

which counts only explicit toxic vocabulary and cannot detect subtler forms of hostility such as sarcasm, or implied threats.

5.2 H2 – Negativity Bias

Board meeting comments are strongly positive in sentiment (mean AFINN score 0.156; 65.4% labelled positive by Bing Liu), while online comments' sentiment shares are distributed evenly across all three labels (mean AFINN 0.033–0.049). Figure 2 shows the sentiment label proportions by source for Bing Liu. In Appendix A, we provide an additional figure that displays the full sentiment distributions for both AFINN and Bing Liu lexicons.

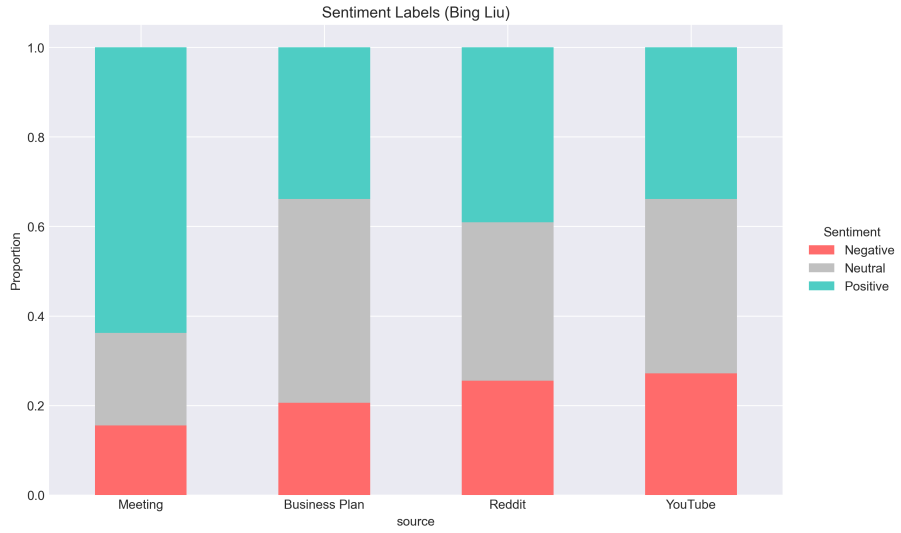


Figure 2: Sentiment label proportions by source (Bing Liu). Board meetings are 65.4% positive; YouTube is 27.2% negative with only 33.8% positive.

To formally test these observations, Table 3 presents Mann-Whitney U test results for both lexicons. All comparisons are one-sided (H_A: online < offline sentiment).

Table 3: H2 – Negativity Bias: Mann-Whitney U test results

Lexicon	Comparison	Online mean	Meeting mean	p-value	Effect size (r)
AFINN	YouTube vs Meeting	0.049	0.156	$p < 10^{-54}$	0.353
	Reddit vs Meeting	0.033	0.156	$p < 10^{-61}$	0.381
	Business Plan vs Meeting	0.034	0.156	$p < 10^{-17}$	0.346
Bing Liu	YouTube vs Meeting	0.056	0.359	$p < 10^{-34}$	0.272
	Reddit vs Meeting	0.107	0.359	$p < 10^{-24}$	0.230
	Business Plan vs Meeting	0.080	0.359	$p < 10^{-15}$	0.316

All six comparisons are highly significant and directionally consistent across both lexicons, with medium effect sizes ($r = 0.23$–0.38). Notably, business plan submissions are significantly more negative than board meeting comments across both lexicons ($p < 10^{-15}$, $r \approx 0.33$), despite both being formal offline channels. This suggests that the act of writing a formal submission, which carries lower immediate social accountability than speaking

before a board in person, allows for somewhat more negative expression. The sentiment ordering across sources (Meeting > Business Plan > Reddit ≈ YouTube) mirrors the participation effort and accountability gradient, consistent with the online disinhibition hypothesis.

The cross-lexicon agreement strengthens confidence in the finding: while AFINN and Bing Liu differ in absolute scale, both identify the same rank ordering and the same pattern of statistical significance, ruling out lexicon-specific artefacts as an explanation.

While sentiment lexicons intrinsically contain more negative than positive terms, this class imbalance does not threaten our findings. Because our study design is strictly comparative, any systematic baseline shift applies uniformly across all sources, leaving the relative divergence between platforms mathematically robust.

5.3 H3 – Relevance

The LDA model yields 7 interpretable topics mapping onto substantive themes in the CAHSR debate, as shown in Table 4.

Table 4: LDA topic labels and top terms ($k = 7$)

Topic	Label	Top terms
0	General commentary	train, like, would, get, people, time, think
1	HSR branding	high speed rail, california high speed, waste
2	Project & funding	project, year, billion, cost, funding, construction
3	Geography	valley, city, central valley, station, bay area, fresno
4	Technical & speed	track, mph, brightline, service, plan
5	Routes & stations	san, merced, bakersfield, palmdale, caltrain, gilroy
6	Politics & fiscal	money, state, tax, highway, pay, government

Table 5 presents Mann-Whitney U test results comparing CAHSR relevance scores across sources.

Table 5: H3 – Constructiveness: Mann-Whitney U test results

Comparison	Online mean	Meeting mean	p-value	Effect size (r)
YouTube vs Meeting	0.087	0.154	$p < 10^{-34}$	0.269
Reddit vs Meeting	0.093	0.154	$p < 10^{-24}$	0.232
Business Plan vs Meeting	0.138	0.154	$p = 0.019$	0.082

YouTube and Reddit comments show significantly lower CAHSR relevance than board meeting comments ($p < 10^{-24}$), with small-to-medium effect sizes ($r \approx 0.23$–0.27). Business plan submissions show a nominally significant difference ($p = 0.019$), but this does not survive Bonferroni correction ($\alpha = 0.0056$), consistent with the H1 finding that formal written submissions resemble in-person testimony in their topical focus.

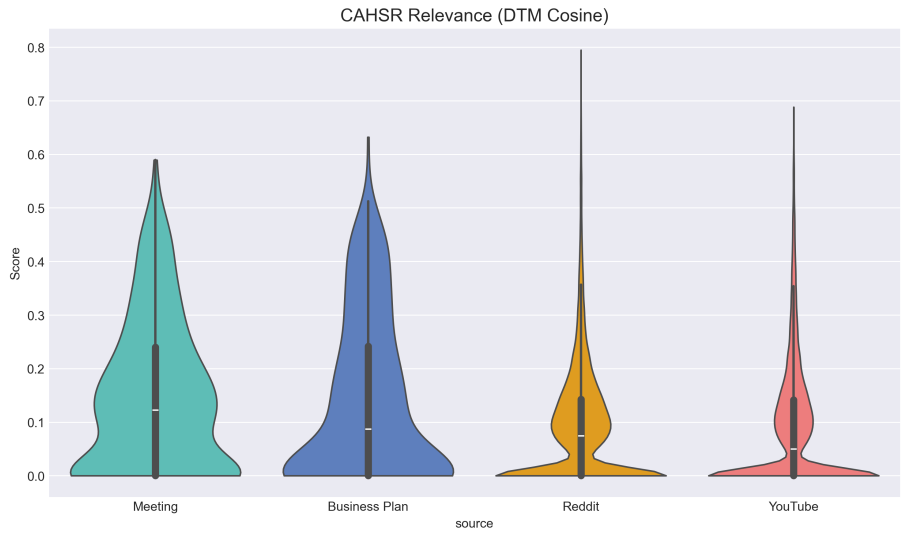


Figure 3: CAHSR relevance score distributions by source (DTM cosine similarity). Board meetings and business plan submissions concentrate at higher relevance scores than online sources.

In Appendix B, we provide an additional figure that displays the topic distribution across sources.

6 Conclusion

This paper set out to test whether the keyboard warrior effect, the online disinhibition phenomenon described by Suler (2004), manifests in public discourse around the California High-Speed Rail project. Motivated by the growing reliance on digital channels for public deliberation, we asked whether these new avenues for civic co-creation provide institutions with the same quality of public signal as traditional, offline channels. Using 62,774 comments from four participation mediums and a battery of NLP methods, we find consistent and statistically robust evidence that they do not. Table 6 summarises the results.

Table 6: Summary of hypothesis test results

H	Dimension	Finding	Effect size	Verdict
H1	Toxicity	Online > Offline	Negligible ($r \approx 0.02$)	Supported
H2	Negativity Bias	Online $\gg$ Offline	Medium ($r > 0.23$)	Strongly supported
H3	Relevance	Online < Offline	Small–medium ($r \approx 0.23$–0.27)	Supported

Explicitly, we found that anonymous online platforms (Reddit and YouTube) are systematically more toxic, significantly more negative, and less topically focused on the specifics of the CAHSR project than in-person board meetings. Online commenters were 12 to 19 times more likely to use hostile language (based on toxicity prevalence rates of 1.9–3.0% online vs. 0.2% offline) and exhibited a strong negativity bias, whereas offline testimony remained overwhelmingly constructive and positive.

A recurring and critical pattern across all three dimensions is the behaviour of the business plan submissions. Despite being submitted through online web portals, they consistently resembled in-person board meeting comments more than social media commentary: they exhibited near-zero toxicity, were significantly less negative than social media, and maintained high topical relevance. This suggests that the participation effort required acts as a self-selection mechanism that filters for constructive participants. Accountability and effort, rather than the digital medium alone, appear to be the true drivers of discourse quality.

These findings carry urgent practical implications for deliberative institutions. As governments increasingly look to the Internet and social media to gauge public sentiment and co-create policy, they must recognize that these channels do not accurately mirror offline civic engagement. If institutions rely heavily on social media commentary as a signal of public opinion, they risk systematically overestimating public hostility and underestimating constructive engagement, as the overwhelming volume of online noise (accounting for 98.5% of our corpus) easily drowns out substantive offline contributions. To prevent the degradation of citizen-institution communication, policymakers must design digital feedback mechanisms that actively require effort and accountability, rather than simply opening the floodgates to anonymous social media.

Limitations. The dictionary-based toxicity method counts only explicit hostile vocabulary and likely underestimates the true extent of hostile language, as evidenced by the negligible absolute effect sizes in H1. The offline sample is substantially smaller than the online sample (936 vs 61,838 comments), which limits statistical power for detecting subtle effects. Furthermore, board meeting attendees are self-selected individuals who invest significant effort to participate in person, introducing a selection bias that partially conflates platform affordances with participant demographics. Finally, the cosine similarity relevance measure depends on the choice of anchor documents, which were constructed manually and have not been independently validated.

Additionally, the temporal coverage of each source differs: YouTube and Reddit span multiple years of organic posting, whereas business plan comments cluster around four biennial consultation windows. The classification of business plan submissions as "offline" reflects the absence of user interaction on that channel, but these submissions are technically web-submitted; the distinction rests on participation cost and accountability rather than transmission medium. Finally, the study is cross-sectional, so causal claims about anonymity driving disinhibition should be treated with caution.

Future work. A natural extension would compare dictionary results against transformer-based classifiers to quantify what proportion of implicit hostility (e.g., sarcasm, condescension) the lexicon approach misses. Temporal event studies linking toxicity and sentiment spikes to specific CAHSR milestones—such as budget overruns, route changes, or funding decisions—would allow for causal identification rather than cross-sectional comparison. Finally, testing this methodological framework on other contested infrastructure projects would assess whether these deliberative breakdowns generalise beyond the CAHSR context.

References

Blei, D. M., Ng, A. Y., and Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of Machine Learning Research*, 3:993–1022.

Brugúe, Q. (2014). Public policy: Between deliberation and the exercise of authority. *Cuadernos de Gobierno y Administración Pública*, 1(1):37–55.

Brugúe, Q. (2022). *Organizaciones que saben, organizaciones que aprenden*. Monografías INAP. Instituto Nacional de Administración Pública, Madrid.

Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, 2nd edition.

Hu, M. and Liu, B. (2004). Mining and summarizing customer reviews. In *Proceedings of KDD*, pages 168–177.

Nielsen, F. Å. (2011). AFINN sentiment lexicon.

Papacharissi, Z. (2004). Democracy online: Civility, politeness, and the democratic potential of online political discussion groups. *New Media & Society*, 6(2):259–283.

Parés, M., Castellà, C., and García Brugada, M. (2015). Repensant la participació de la ciutadania al món local. Documents de Treball. Igualtat i Ciutadania, 21. Diputació de Barcelona, Barcelona.

Santana, A. D. (2014). Virtuous or vitriolic: The effect of anonymity on civility in online newspaper reader comment boards. *Journalism Practice*, 8(1):18–33.

Suler, J. (2004). The online disinhibition effect. *CyberPsychology & Behavior*, 7(3):321–326.

A Additional Figures for H2

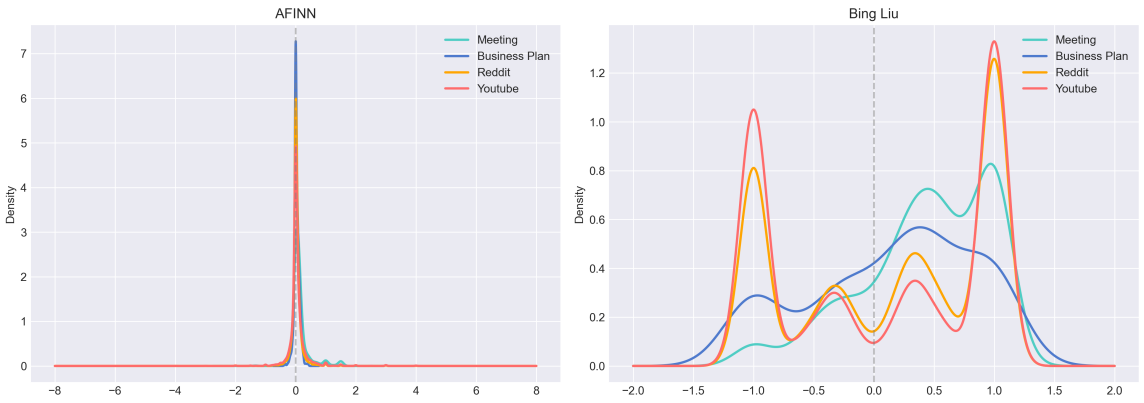


Figure 4: Sentiment score distributions by source: AFINN (left) and Bing Liu compound score (right). Board meeting comments shift markedly toward positive sentiment relative to all online sources.

B Additional Figures for H3

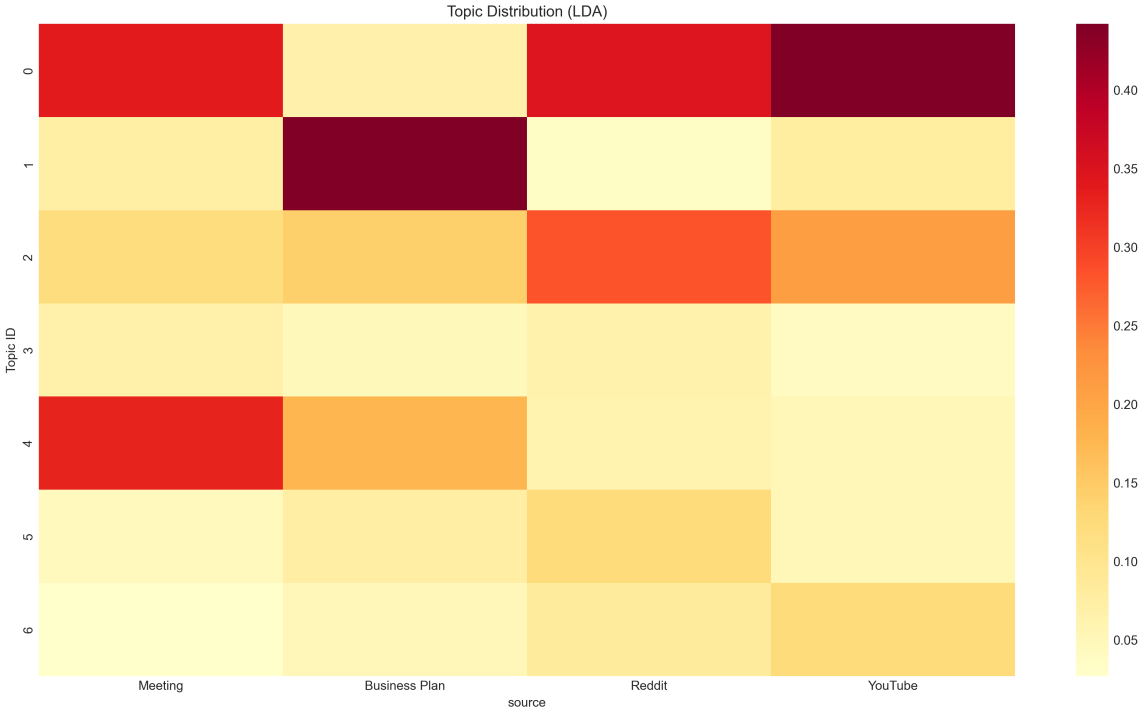


Figure 5: LDA topic distribution by source (column-normalised). Offline sources concentrate on project-specific topics (2, 3, 5); online sources show more diffuse engagement including general commentary (0) and political/fiscal criticism (6).